Lost in Simplification: Detecting Safety-Critical Errors in LLM-Generated Medication Instructions

Rohit Kulkarni
Khoury College of Computer Sciences
Northeastern University
kulkarni.rohita@northeastern.edu

Atharv Talnikar
Khoury College of Computer Sciences
Northeastern University
talnikar.a@northeastern.edu

Christopher Huitt
Khoury College of Computer Sciences
Northeastern University
huitt.c@northeastern.edu

Abstract

Medication non-adherence affects up to 50% of patients with chronic diseases and contributes to an estimated 100,000 preventable deaths annually. A key driver is the complexity of FDA-mandated drug labeling, which is written for clinicians rather than patients. Large language models (LLMs) can rewrite these labels into plain language, but using them for simplification bring the risk of dropping or weakening safety-critical information such as dosage instructions, contraindications, warnings, and drug interactions. We designed an end-to-end pipeline that ingests structured drug labels from the openFDA API, generates simplified versions using two open-weight LLMs (GPT-OSS-120B and Llama-3.3-70B via Groq), and systematically evaluates safety preservation through heuristic comparison and LLM-as-judge evaluation. Across 25 medications spanning three risk categories, we find that only 38.7% of safety-critical fields are fully preserved after simplification. Drug interactions and contraindications are the most vulnerable, with drop rates exceeding 81%. High-risk and controlled substances fare worst, with only 28.9% of fields preserved compared to 65.4% for over-the-counter drugs. Manual validation of the LLM judge yields 60% agreement ($\kappa = 0.27$). Readability improves by an average of 2.6–4.1 Flesch–Kincaid grade levels, but gains do not correlate with better safety preservation, suggesting that readability metrics alone are insufficient for evaluating LLM-based medical text simplification.

1 Introduction

The adverse effects of medication nonadherence, which is the failure to follow the directions provided with your medications, have been demonstrated to affect up to 50% of patients suffering from chronic illnesses in the United States [Brown and Bussell, 2011]. Adverse effects include serious consequences: estimates indicate that approximately 125,000 deaths could be prevented each year if patients followed their medication orders, and the cost to provide care to those who do not would be avoided by approximately $100 billion dollars per year [Viswanathan et al., 2012]. Although there are numerous contributing factors associated with why patients fail to adhere to their medication orders, a major factor has been shown to be the complexities of medication packaging. The U.S. Food and Drug Administration (FDA) requires that all "package insert" material for over-the-counter (OTC) analgesics such as ibuprofen, through Schedule II controlled substances such as oxycodone, be prepared for use by clinical professionals and regulatory officials [FDA, 2006]. Frequently, however,

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patients are treated as an after-thought and must interpret dense language describing concepts that assume medical education/training they have not received.

Modern generative large language models (LLMs) appear to present a viable option for addressing this issue. For example, modern generative LLMs can translate complex technical writing into simple, clear language understandable by laypersons in seconds, which may assist in closing the health literacy gap for millions of patients. Simplifying technical information is inherently a dangerous process in safety-related areas; i.e. the act of translating complex writing into simpler writing may intentionally or unintentionally cause changes to the intended meaning of the original text. Therefore, if an LLM reduces the emphasis of a contraindication, deletes a dosing restriction, deletes a black-box warning, or deletes a potential drug-drug interaction, then the resulting simplified version of the text may result in actual harm to the patient (including but not limited to life threatening injury/death).

Research on LLMs used to generate summaries has concentrated on accuracy metrics [Lin, 2004, Zhang et al., 2020]. While useful for measuring surface-level similarity, these metrics are insensitive to the kinds of errors that matter most in healthcare: a simplification that scores well on semantic similarity can still omit a critical warning or alter a dosage instruction. To our knowledge, no prior work has systematically characterized the types and frequencies of safety-critical errors that LLMs introduce when simplifying medication instructions.

In this paper, we address this gap by presenting an end-to-end pipeline for detecting adherence-critical errors in LLM-generated medication simplifications. Our contributions are as follows:

1. We design a structural testing framework for evaluating adherence-relevant errors introduced when simplifying medication instructions using two weight-based LLMs. The framework uses data from the FDA obtained via the openFDA API and consists of four steps: first, obtain FDA-approved label data via the openFDA API; second, use the two weight-based LLMs to produce simplified versions of the label data; third, extract five safety-critical fields from the original and simplified label data; fourth, use both rule-based heuristics and an LLM-as-judge approach using a three-category taxonomy (Preserved, Softened, Dropped) to determine whether each field was preserved.
2. We utilize the pipeline to evaluate adherence-relevant error generation across 25 medications that fall within three clinically significant risk categories (OTC medications, common prescription medications, and medications classified as high-risk/control). In total, we found that less than half of the safety-critical fields extracted from the original label data survived simplification without being altered.
3. We analyze the relationship between improvements in readability and safety content preservation. We demonstrate that these goals are inversely related: medications with the greatest improvement in readability had significantly greater losses in safety content.
4. We evaluate the inter-rater reliability between ratings assigned by the LLM judge and ratings assigned by human raters. This provides us with a means to calibrate the judge's strengths and systematic biases.

2 Related work

Health literacy and medication comprehension. Health literacy is an individual's ability to gather, understand, and apply medical information. It is one of the most important factors influencing medication adherence [Berkman et al., 2011]. In fact, approximately 36% of adult Americans in the United States have either basic or no health literacy [Kutner et al., 2006]; and those who do experience lower levels of health literacy are also at risk of being unable to comprehend their medication instructions, having higher rates of hospitalizations, and experiencing higher mortality rates than individuals with adequate health literacy [DeWalt et al., 2004]. The FDA has recognized the issue of health literacy as a problem by implementing the "Physician Labeling Rule" [FDA, 2006]; however, it appears that labeling still remains at a college reading level and does not meet the federal recommendation for labeling patients' materials, which should be at least in the range of 6th – 8th grade [Weiss, 2007].

Text simplification in healthcare. Automated text reduction has been studied across a large spectrum of natural language processing (NLP) [Zhang and Lapata, 2017, Alva-Manchego et al.,

2020], as well as specifically within biomedical research areas [Van den Bercken et al., 2020, Devaraj et al., 2021]. The early work using automated text simplification utilized a variety of rules and statistical methods. More recent work utilizing large language models demonstrated high levels of accuracy when evaluated against standard benchmarking for automatic text simplification [Feng et al., 2023]. Medical literature has also seen a number of articles that apply large language models to automatically reduce radiology and discharge report complexity [Lyu et al., 2023, Jeblick et al., 2024, Zaretsky et al., 2024]. Those studies primarily used established readability metrics (SMOG and Flesch-Kincaid) along with general NLP metrics (ROUGE, BERTScore) in evaluating their results. These are all typically surface level measures and may indicate how readable the resulting reduced version is. However, they do not assess whether critical elements of patient care were lost during this process.

LLM evaluation and factual consistency. The use of large language models (LLMs) as evaluators ("the LLM-as-judge" model), has been gaining momentum as a viable option to replace human evaluation [Zheng et al., 2024]. Prior research demonstrates that LLM judges are capable of approximating human preferences for open-ended generation tasks; however, these models also demonstrate systemic biases such as position bias and verbosity preference [Zheng et al., 2024]. For example, in healthcare, some research has demonstrated the potential utility of using LLMs for factual consistency checking in the context of clinical summary creation [Tang et al., 2023], but this type of application has yet to be applied in the specific case of creating simplified versions of medication labels. Since the error taxonomy associated with the task of labeling medications (i.e. dropped warnings, softened contraindications, altered dosing), is both domain- and clinically significant, it will likely require a high level of accuracy for each individual item on the taxonomy to ensure safe patient outcomes.

Drug label data and the openFDA API. The openFDA Drug Label API [openFDA, 2024] provides an organized way of accessing the FDA approved label for medications, which includes all of the required information in a machine readable format (e.g., dosing, warnings, contraindications, drug interactions, and boxed warnings). This provides a structural reference point for comparing simplification outputs with well defined ground truths. It provides an important advantage in comparison to other areas where evaluation depends on subjective human judgment.

3 Methodology

Our pipeline consists of five stages: (1) data ingestion from the openFDA API, (2) LLM-based simplification, (3) structured safety field extraction, (4) heuristic comparison, and (5) LLM-as-judge semantic evaluation. Figure 1 provides an overview.

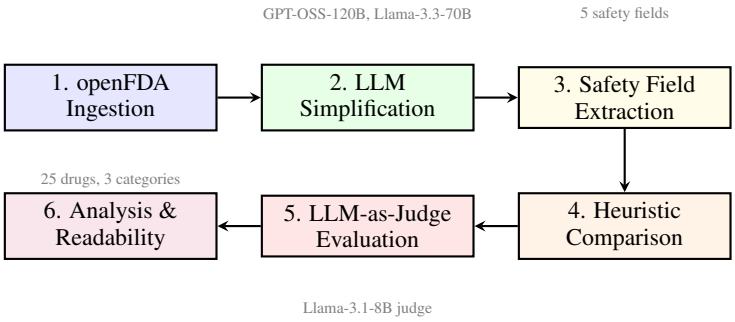

Figure 1: End-to-end pipeline architecture. Raw FDA labels are ingested, simplified by two LLMs, decomposed into five safety-critical fields, and evaluated through both heuristic and semantic comparison.

3.1 Dataset construction

We curated a dataset of 25 medications that spanned three clinically relevant risk-based categories: **Simple OTC** (8 drugs), **Common Prescription** (8 drugs), and **High-Risk/Controlled** (9 drugs). Table 1 lists all the medicines included in this study, along with their categories. The categories were

chosen to provide a range of differences in dosing complexity, the seriousness of interactions, and patient risk. OTC products generally have less complex labeling and are used by larger populations than prescription products. Commonly prescribed products present more moderate levels of complexity than both simple OTC and high risk/controlled products. High-risk/controlled products contain boxed warnings, have very narrow therapeutic indices, or possess the potential for abuse.

Labels for all of these drugs were retrieved from the openFDA Drug Label API [openFDA, 2024] using generic names for each drug. If there was more than one label available for any given drug, we chose the label that contained the greatest number of safety-critical fields as well as the most recently effective date. All raw labels underwent decodification of HTML entities, removal of artifact section numbering, and normalization of all whitespace.

Table 1: Dataset of 25 medications across three risk categories.

Category	Medications
Simple OTC (8)	Ibuprofen, Acetaminophen, Diphenhydramine, Loratadine, Omeprazole, Cetirizine, Aspirin, Naproxen
Common Rx (8)	Amoxicillin, Metformin, Lisinopril, Atorvastatin, Metoprolol, Sertraline, Prednisone, Azithromycin
High-Risk / Controlled (9)	Methotrexate, Tofacitinib, Warfarin, Isotretinoin, Clozapine, Oxycodone, Alprazolam, Amphetamine, Diazepam

3.2 LLM-based simplification

We developed simplified medication instructions using the two open-weight llms hosted on the Groq inference platform: **GPT-OSS-120B** (`openai/gpt-oss-120b`) and **Llama-3.3-70B-Versatile** (`llama-3.3-70b-versatile`). The two models were each prompted using a zero-shot approach at a temperature of 0.2 to promote cautious and accurate reductions of complexity.

The prompt design focused on preserving safety by including several safety constraints in the user prompt. The system prompt instructed the model to act as a medical instruction simplification assistant that "*MUST preserve adherence- and safety-critical information exactly or with equivalent meaning.*" The user prompt included eight specific safety constraints: (1) do not remove dosage information; (2) do not remove or soften warnings; (3) do not remove or weaken contraindications; (4) do not omit drug interactions; (5) keep all dose amounts, units, and frequencies unchanged; (6) do not invent new warnings; (7) do not soften language (e.g., do not replace "contraindicated" with "not recommended"); and (8) return valid JSON with structured fields.

The model received a hybrid input consisting of both structured safety fields (boxed warnings, dosage, warnings, contraindications, interactions) with shortened original label text where structured fields did not exist. The requested output consisted of a JSON object with six keys: `boxed_warning`, `dosage`, `warnings`, `contraindications`, `interactions`, and `simplified_text`. Due to Groq's free tier token limit of approximately 7,600 tokens available per request, the pipeline adjusted the maximum number of output tokens based on the length of the prompts. Also included is a fallback to plain text parsing when generating JSON fails.

3.3 Safety field extraction

To allow for a methodical comparison, we identified five fields within the safety-related areas of the original and simplified label formats. These fields were then standardized to ensure each field was represented in a consistent manner, regardless of the original format. The five fields included: (1) **Boxed warning**: the most serious FDA-mandated safety alert; (2) **Dosage and administration**: dose amounts, frequencies, routes, special instructions, etc.; (3) **Warnings**: general warnings about adverse effects and precautions; (4) **Contraindications**: conditions or populations for which the drug should not be used; and (5) **Drug interactions**: known interactions with other drugs, foods, or substances.

The mapping for original labels came directly from the structured API response provided by OpenFDA. For example, "*drug_interactions*" was mapped to "*interactions*". The simplified labels were also extracted from the JSON format returned by the LLM. A text-based fallback parser was then also

utilized for cases where the model returned unstructured text instead of JSON. Any empty/null fields were logged and not silently discarded so that any downstream comparisons can differentiate between missing content and failed extractions.

3.4 Heuristic comparison

The first evaluation layer applied three rule-based heuristics to flag potential safety-content loss:

1. **Dropped field**: The original label contains text for a safety field, but the simplified version is empty.
2. **Possible dose mismatch**: Numeric dose tokens (units such as “mg” or “mcg,” and frequency markers such as “q12h” or “bid”) extracted from the original dosage field by regular expressions are absent from the simplified version.
3. **Possible content loss**: For non-dosage fields, the first 200 characters of the original (after normalization) do not appear as a substring in the simplified text, and at least one of three significant sentences (>40 characters) is absent.

These heuristics were designed for high recall. They cast a wide net to ensure no safety-critical loss goes undetected, at the cost of precision. Across our dataset, 104 issue flags were raised for 25 drugs, with every drug having at least one flag.

3.5 LLM-as-judge evaluation

The second layer of evaluation was designed to address the semantic limitations of heuristic string matching with the use of a judge model, **Llama-3.1-8B-Instant** (via Groq), to evaluate the simplified safety fields compared to their original counterparts. Using the prompting feature, the judge model would be asked to compare the five safety fields for both the original labels and the simplified labels and then asked to assign one of three verdicts to each comparison:

- **Preserved**: All safety-critical information from the original label is present in the simplified version, even if reworded.
- **Softened**: Core safety information is present but its urgency or specificity has been weakened (e.g., "contraindicated” becomes "not recommended,” or specific drug names removed from an interaction list).
- **Dropped**: Safety-critical information is missing, contradicted, or replaced with filler text.

The judge evaluated each field independently, receiving the full text of the original and simplified field as input. When the simplified field was empty but the original contained content, the verdict was automatically set to Dropped without invoking the LLM. When both fields were empty, the pair was skipped. The judge operated at temperature 0.2 with a maximum of 768 output tokens per call.

3.6 Manual validation

A team member manually labeled 20 drug pairings to evaluate the performance of the LLM judge. Each drug pairing was evaluated using a three category outcome (Preserved, Softened, or Dropped). The evaluations were compared with the LLM judge’s output to determine the percentage of agreement and to calculate Cohen’s Kappa (κ) for multi-class classification. The 20 drug pairings represented all 5 safety fields and a wide variety of drugs. The manual validation was conducted on an initial pipeline run; the judge model and evaluation procedure were identical to the primary analysis reported here.

3.7 Readability assessment

We measured readability on both original and simplified texts using four standard indices computed via the `textstat` Python library: Flesch Reading Ease (FRE; 0–100 scale, higher is easier), Flesch–Kincaid Grade Level (FK; U.S. school grade), SMOG Index (grade level based on polysyllabic words), and Gunning Fog Index (grade level based on sentence complexity). Metrics were computed per drug, and deltas were used to quantify improvement. Text length in characters was recorded to measure compression.

4 Results

We present the results from the LLM-as-judge evaluation of the GPT-OSS-120B simplification, our primary analysis, followed by a cross-model readability comparison with Llama-3.3-70B.

4.1 Safety preservation by field

Table 2 reports the judge's results aggregated by safety field. Of the 106 total judgments across all fields, only 41 (38.7%) were rated as Preserved. The vulnerability varies sharply by field. Boxed warnings had a 92.9% preservation rate, with only one instance of a dropped boxed warning. Dosage information was Preserved 64.0% of the time, but 36.0% of dosage fields were Dropped outright with no instances of softening. Contraindications and drug interactions were the most vulnerable. Contraindications were Dropped in 81.8% of cases with no instances of softening—the model either preserved them or omitted them entirely. Drug interactions fared worst overall: not a single interaction field was fully Preserved, and 90.5% were Dropped.

Table 2: Safety preservation rates by field across 25 drugs (LLM judge, GPT-OSS-120B simplification). Drug interactions and contraindications are the most vulnerable fields.

Field	Judged	Preserved (%)	Softened (%)	Dropped (%)
Boxed Warning	14	13 (92.9)	0 (0.0)	1 (7.1)
Dosage	25	16 (64.0)	0 (0.0)	9 (36.0)
Warnings	24	8 (33.3)	5 (20.8)	11 (45.8)
Contraindications	22	4 (18.2)	0 (0.0)	18 (81.8)
Drug Interactions	21	0 (0.0)	2 (9.5)	19 (90.5)
Overall	**106**	**41 (38.7)**	**7 (6.6)**	**58 (54.7)**

4.2 Safety preservation by drug category

Table 3 shows how preservation rates differ based on the three drug categories. Preservation percentage is highest for Simple OTC drugs at 65.4%. This may be due to the fact that the labels used for this type of drug are generally shorter in length than other types of medication labels. They also tend to have fewer fields and use simpler forms of clinical language. Common Rx drugs had a substantially lower preservation rate of 31.4%, with 68.6% of fields Dropped as the model attempted to simplify more nuanced clinical text. High-risk and controlled substances fared worst overall, with only 28.9% Preserved and 57.8% categorized as Dropped. These figures reflect the difficulty associated with reducing the large number of words contained in the typical warning labels for these types of medications. These labels can include many warnings such as those regarding dangerous side effects, drug interactions, and dosage instructions.

Table 3: Safety preservation rates by drug risk category. High-risk drugs fare substantially worse than OTC drugs.

Category	Drugs	Judged	Preserved (%)	Softened (%)	Dropped (%)
Simple OTC	8	26	17 (65.4)	1 (3.8)	8 (30.8)
Common Rx	8	35	11 (31.4)	0 (0.0)	24 (68.6)
High-Risk/Controlled*	9	45	13 (28.9)	6 (13.3)	26 (57.8)

*Includes one combination product (dextroamphetamine/amphetamine) assigned to this category for reporting.

4.3 Heuristic versus LLM judge agreement

Table 4 compares the heuristic flags against the LLM judge's verdicts. The heuristic layer achieved perfect recall. There were zero cases where the judge identified a softened or dropped field that the heuristic had not flagged. However, precision was low with 39 of 104 heuristic flags (37.5%) being false alarms, where the judge deemed the content Preserved despite a heuristic flag. This confirms the intended design. The heuristic layer is a high-recall, low-precision first pass that motivates the more expensive semantic evaluation.

Table 4: Heuristic comparison vs. LLM judge agreement (106 field pairs). Heuristics catch all issues but produce 37.5% false alarms.

	Judge: Softened/Dropped	**Judge: Preserved**
Heuristic flagged	65 (agreement)	39 (false alarm)
Heuristic not flagged	0 (miss)	2 (both clear)

Heuristic sensitivity: 100%. Heuristic false alarm rate: 37.5%.

4.4 Manual validation of the LLM judge

Manual annotation of 20 drug-field pairs shows that there was an overall 60.0% level of agreement between human annotations and the LLM judge in terms of drug-field pair classification. Cohen's kappa was found to be $\kappa = 0.27$, which can be classified as fair agreement. Table 5 presents the confusion matrix. The most common pattern of disagreement was the judge classifying a field as Dropped when the human annotator labeled it Softened (5 of 8 disagreements). This suggests the judge is systematically stricter than human raters on cases of partial information loss as it tends to treat partial omissions as complete drops rather than recognizing that some safety content was preserved in weakened form.

Two other examples of false Preserved classifications arose. The prescriber and patient alert/warning on Ibuprofen and the boxed warning on clozapine were classified as Preserved by the judge. However, both were classified Softened or Dropped by the human annotator after a closer review. Both examples relied upon omission of detail in the simplified text, specifically omitting particular clinical information (e.g., clozapine's neutropenia monitoring).

Table 5: Confusion matrix: human annotations vs. LLM judge (20 pairs). Rows are human labels; columns are judge verdicts. The judge tends to over-classify Softened as Dropped.

	Judge: Preserved	**Judge: Softened**	**Judge: Dropped**
Human: Preserved	0	0	0
Human: Softened	1	4	5
Human: Dropped	1	1	8

Agreement: 60.0%. Cohen's $\kappa = 0.27$ (fair). Judge over-classifies Softened as Dropped.

4.5 Readability analysis

Table 6 compares the readability of the two different versions of labels that were created for both models, specifically the original version of the label and the simplified version of the label. The greatest improvements in terms of readability were made using the GPT-OSS-120B model. It achieved the largest improvement in readability by reducing the Flesch–Kincaid grade level by an average of 4.1 points from 14.3 to 10.2. It also improved the Flesch Reading Ease by 19.5 points on average. While still showing some readability improvements, the readability improvements made with the Llama-3.3-70B model were less than those made with the GPT-OSS-120B model. It reduced the FK grade by 2.6, and seven drugs actually received *worse* readability scores after simplification, compared to two for GPT-OSS.

Both models achieved extreme text compression. The original labels from openFDA averaged 78,114 characters, while GPT-OSS simplifications averaged 1,103 characters. This resulted in a 98.6% reduction. Llama simplifications averaged out to 1,016 characters. This resulted in a 98.7% reduction. This compression ratio of roughly 70:1 is inherent to the task, as full package inserts contain extensive clinical detail that a patient-facing summary necessarily condenses.

Table 6: Average readability metrics: original vs. simplified across 25 drugs. GPT-OSS achieves larger readability improvements but both models dramatically compress the source text.

	GPT-OSS-120B			Llama-3.3-70B		
Metric	**Orig.**	**Simp.**	**Δ**	**Orig.**	**Simp.**	**Δ**
Flesch Reading Ease (↑)	28.9	48.3	+19.5	28.9	39.7	+10.8
FK Grade Level (↓)	14.3	10.2	−4.1	14.3	11.7	−2.6
SMOG Index (↓)	15.5	12.0	−3.5	15.5	13.3	−2.2
Gunning Fog (↓)	17.4	12.3	−5.0	17.4	14.2	−3.1
Avg. chars	78,114 → 1,103			78,114 → 1,016		
Worse readability (# drugs)		2			7	

4.6 Readability–safety tradeoff

Figure 2 plots the relationship between readability improvement (Flesch–Kincaid grade level reduction) and safety preservation rate (proportion of fields judged Preserved) for each drug. Contrary to what might be expected, there is no positive correlation between the two: drugs with the largest readability gains did not exhibit better safety preservation. Prednisone, for example, achieved one of the largest readability improvements (10.1 FK grade levels) but had only 1 of 4 safety fields preserved. Conversely, several OTC drugs with minimal readability change (1–2 grade levels) achieved 100% preservation.

The implications of this study are as follows: Readability metrics do not adequately determine the quality of LLM-based medication simplification. Although an LLM based system generates very readable results, it may produce the largest and most serious safety omissions.

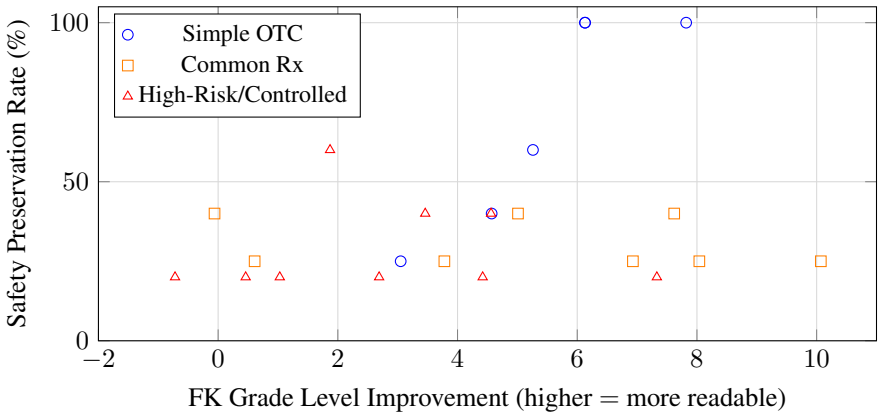

Figure 2: Readability improvement (FK grade level reduction) vs. safety preservation rate per drug, colored by risk category. No positive correlation is observed; the largest readability gains often co-occur with low safety preservation.

5 Discussion

Why interactions and contraindications fail. The stark vulnerability of drug interactions (90.5% dropped) and contraindications (81.8% dropped) likely reflects two compounding factors. First, these sections in original FDA labels tend to be long, list-heavy, and clinically dense. Interactions sections often enumerate dozens of specific drug-drug pairs with varying severity levels. When the LLM compresses a 78,000-character label into approximately 1,100 characters, these lengthy lists are among the first to be affected. Second, the model we used may lack sufficient training signal to distinguish safety-critical interactions from background clinical detail. This leads it to treat all content as equally compressible. The binary pattern in contraindications—18.2% preserved, 81.8% dropped, and 0% softened—tells us that the model either reproduces the contraindication list faithfully or omits it entirely. There is no middle ground.

Why boxed warnings succeed. Boxed warnings achieved 92.9% preservation. We believe that the reason for this was due to how Boxed Warnings were presented in our training data. Boxed Warnings are typically short, prominent, and have an obvious indicator of importance ("the black box") that signals the reader that what they are reading is very critical. Also, being shorter allows them to be preserved during compression much better than longer sections.

The readability–safety disconnect. Our scatter analysis (Figure 2) shows that readability improvement and safety preservation are independent. This finding challenges the assumption in prior work that better readability scores indicate better simplification quality. Prednisone illustrates the danger. It achieved one of the highest readability improvements in our dataset while having three out of four safety fields dropped. For a patient reading the simplified output, the text would appear clear, well-organized, and easy to understand, but it had omitted critical contraindications, drug interactions, and warnings. This argues strongly for safety-specific evaluation metrics in any deployment of LLM-based medication simplification.

LLM judge calibration. The 60% agreement rate and $\kappa = 0.27$ between the LLM judge and human annotations represent fair yet not perfect level of reliability. The judge's systematic bias toward over-classifying Softened cases as Dropped (Table 5) means our reported drop rates may be inflated some. However, this conservative bias is arguably preferable in a safety-critical context: it is safer to over-flag potential losses than to miss them. The two false Preserved cases (ibuprofen and clozapine boxed warnings) are more concerning, as they represent instances where the judge failed to detect real safety-content loss. These cases involved subtle omissions of specific clinical details within an otherwise preserved general warning structure. This shows the judge struggles with partial losses in fields it recognizes as present.

Cross-model comparison. The GPT-OSS-120B model produced larger readability changes than did Llama-3.3-70B. The average readability improvement was 4.1 vs. 2.6 FK-grade levels, and there were fewer readability degradations in GPT-OSS (2 drugs vs. 7 in Llama). However, when both models were evaluated by the same LLM judge, Llama preserved 43.4% of safety fields compared to 38.7% for GPT-OSS. While GPT-OSS achieves larger readability gains, it does not translate into greater safety preservation. The Llama model's more conservative simplifications—with less dramatic output length reductions and smaller readability deltas—appear to retain slightly more safety content, suggesting a trade-off between compression aggressiveness and fidelity.

6 Limitations

Our study has several important limitations. The first limitation of our research is that we used a relatively small model for the LLM Judge (Llama-3.1-8B-Instant), so although it made judgments about how accurate the LLM was at identifying lost information, those judgments have not been validated by clinical pharmacists nor domain experts.The κ statistic of 0.27 shows good agreement with what we manually annotated, there is still much opportunity for disagreement among judge. Also, the judge's systematic bias of classifying partial losses as Dropped could have inflated our estimated error rate.

Another limitation is that our sample size of 25 drugs is very small compared to the over 100,000 drugs approved by the FDA. It is possible that results will not generalize well to other types of drugs not well-represented within our sample (for example, biologics, combination therapies, pediatric formulations).

Also, we relied solely upon the openFDA API for ground truth. We do know that not all of the labels returned from the API are equally complete. For example, many OTC products lack detailed contraindications/interactions sections within their API response.

A further limitation is that our heuristic comparison layer depends upon string matching, which cannot determine if two strings are semantically equivalent. This issue has been somewhat mitigated by use of the LLM judge, however, it is an issue related to using a two stage design.

Lastly, we only tested zero-shot prompting in our research. Techniques like few-shot prompting, chain-of-thought reasoning, or retrieval-augmented generation (RAG) could potentially provide better safety preservation and were not researched.

7 Conclusion

We demonstrated a complete pipeline from input (the original drug label) through output (a simplified version of the label created by the LLM), with the objective of identifying safety-critical errors in the LLM's versions of the drug labels. The results of evaluating twenty-five drugs across three different levels of risk show that there is a significant chance that current LLMs will either omit or diminish critical adherence-related information in the safety fields. The two highest risk areas identified were drug interactions and contraindications. Finally, high-risk drugs performed worse than both medium- and low-risk drugs. Furthermore, we found no relationship between improved readability and safety preservation. Overall, these findings demonstrate that the use of LLM to create medication labels that have been simplified poses unacceptable risks to patients' safety, absent additional measures to verify their accuracy.

Our analysis also has several potential avenues for research. In particular, we recommend that clinical pharmacists verify the accuracy of the outputs generated by the pipeline developed here, in order to establish a baseline "gold standard" against which the accuracy of the LLM can be evaluated. Next, we recommend expanding the pool of drugs to be tested to better represent the diversity of medications available. There may also be alternative approaches to prompting the LLM to generate safer versions of medication labels; for example, few-shot learning experiments, experiments demonstrating the benefits of using chain-of-thought reasoning, and experiments demonstrating the benefits of using retrieval augmented generation. Additionally, it would appear that multi-model ensemble evaluations would provide one way to mitigate the impact of bias present in the use of a single model to evaluate the outputs generated by another. Finally, human included workflows in which clinicians review LLM-generated simplifications before they reach patients represent a practical path toward safer deployment.

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